Advanced Excel — Student Handout

Batch: 2nd Year MBA · **Session:** Advanced Excel for Business, Day 2 · **Duration:** 3 Hrs

This handout covers everything from today's session — definitions, worked examples, function reference tables, and the hands-on exercises — so you can revise it later without depending on your notes.

Datasets for today. Two files work together throughout this session:

- `employee_master.csv` — a 40-row employee reference table. Columns: `Emp ID, First Name, Last Name, Department, Designation, Hire Date, Base Salary, Bonus %, Region, Manager, Performance Rating, Email, Phone, Status`
- `monthly_transactions.csv` — a 180-row expense transaction log. Columns: `Txn ID, Emp ID, Transaction Date, Category, Amount, Payment Mode, Approved By, Remarks, Project Code`

The transactions reference employees by Emp ID — you'll use lookup functions to pull employee details into the transaction sheet, then aggregate and pivot the combined data.

Learning Objectives

By the end of today you should be able to:

- Use VLOOKUP, XLOOKUP, and INDEX-MATCH to pull data from a reference table
- Aggregate data conditionally with SUMIFS, COUNTIFS, and AVERAGEIFS
- Manipulate text and dates with built-in functions
- Handle errors defensively with IFERROR, IFNA, and ISERROR
- Build PivotTables and PivotCharts to summarise and visualise data interactively

Today's Agenda

TIME	TOPIC
0:00 – 0:10	Recap & why lookups replace nested IF
0:10 – 0:35	VLOOKUP & XLOOKUP
0:35 – 0:55	INDEX-MATCH
0:55 – 1:05	Break
1:05 – 1:30	SUMIFS, COUNTIFS, AVERAGEIFS
1:30 – 1:50	Text & date functions
1:50 – 2:05	Error handling
2:05 – 2:15	Break
2:15 – 2:50	PivotTables & PivotCharts
2:50 – 3:00	Recap & exit ticket

1. Why Lookups Replace Nested IF

In Day 1 we wrote nested IFs to assign regional targets:

```
=IF(C2="North",180000,IF(C2="South",150000,IF(C2="East",200000,165000)))
```

This works for 4 regions but becomes unreadable at 6 or 10. A **lookup function** stores the values in a separate reference table and lets the formula look them up by key — cleaner, scalable, and far easier to update.

2. VLOOKUP

Definition: Searches the **first column** of a table range for a value and returns a value from a specified column in the same row.

```
=VLOOKUP(lookup_value, table_array, col_index_num, [range_lookup])
```

ARGUMENT	MEANING
`lookup_value`	The key to search for (e.g. `"E1001"`)
`table_array`	The reference table range — first column must contain the keys
`col_index_num`	Which column of the table to return (1 = first, 2 = second, etc.)
`range_lookup`	`FALSE` for exact match (almost always); `TRUE` for approximate match

Example: Pull the **Department** for a transaction's employee:

```
=VLOOKUP(B2, employee_master!$A$2:$N$41, 4, FALSE)
```

- `B2` — the Emp ID in the transaction row
- `$A$2:$N$41` — absolute-referenced so copying down doesn't shift the range
- `4` — column 4 = Department
- `FALSE` — exact match

Three VLOOKUP limitations:

1. The lookup key must be in the **first column** of `table_array` — no looking to the left
2. `col_index_num` is a hardcoded number — insert a column and it silently returns the wrong data
3. Only returns the **first** match

3. XLOOKUP (Excel 365 / 2021+)

Definition: The modern replacement for VLOOKUP — no first-column restriction, no fragile column index, and a built-in error message.

```
=XLOOKUP(lookup_value, lookup_array, return_array, [if_not_found], [match_mode], [searc
```

ARGUMENT	MEANING
`lookup_value`	The key to search for
`lookup_array`	The column to search in (any column — not restricted to the first)
`return_array`	The column to return from (can be to the left of the lookup column)
`if_not_found`	Custom message if no match (replaces the need for IFERROR)
`match_mode`	`0` = exact (default), `-1` = exact or next smaller, `1` = exact or next larger, `2` = wildcard
`search_mode`	`1` = first-to-last (default), `-1` = last-to-first

Example: Same Department lookup:

```
=XLOOKUP(B2, employee_master!$A$2:$A$41, employee_master!$D$2:$D$41, "Not Found")
```

No column index. Lookup and return columns are specified independently. `"Not Found"` handles missing IDs without a separate error wrapper.

Example — left-direction lookup (VLOOKUP can't do this): Given an Email, find the Emp ID:

```
=XLOOKUP(L2, employee_master!$L$2:$L$41, employee_master!$A$2:$A$41, "Not Found")
```

4. INDEX-MATCH

Definition: A two-function combination that works in every version of Excel. `MATCH` finds the **position** of a value; `INDEX` returns the **value** at that position.

MATCH

```
=MATCH(lookup_value, lookup_array, [match_type])
```

- `match_type`: `0` = exact match, `1` = largest ≤ value (ascending data), `-1` = smallest ≥ value (descending data)

```
=MATCH("E1010", employee_master!$A$2:$A$41, 0)   → 10
```

INDEX

```
=INDEX(return_array, row_num)
```

```
=INDEX(employee_master!$D$2:$D$41, 10)   → "IT"
```

Combined

```
=INDEX(employee_master!$D$2:$D$41, MATCH(B2, employee_master!$A$2:$A$41, 0))
```

"Find where `B2`'s value sits in the Emp ID column, then return that same row's value from the Department column."

Example — pull the Manager:

```
=INDEX(employee_master!$J$2:$J$41, MATCH(B2, employee_master!$A$2:$A$41, 0))
```

When to use which

NEED	USE
Simple exact lookup, key in first column	VLOOKUP (FALSE)
Lookup from any column, built-in error	XLOOKUP (Excel 365+)
Works everywhere, any direction	INDEX-MATCH
Approximate match (tax brackets, grades)	VLOOKUP (TRUE) or XLOOKUP with match_mode -1

Break (10 min)

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5. SUMIFS, COUNTIFS, AVERAGEIFS

Definition: Conditional aggregation functions — they answer questions like "total travel spend by the Finance department" without filtering the data first.

SUMIFS

```
=SUMIFS(sum_range, criteria_range1, criteria1, [criteria_range2, criteria2], ...)
```

Adds up all values in `sum_range` where **every** criterion is satisfied simultaneously. The sum range comes **first**, then criteria pairs.

Example — total Travel spend:

```
=SUMIFS(E2:E181, D2:D181, "Travel")
```

Example — Travel spend in Q1 2026 (Jan–Mar):

```
=SUMIFS(E2:E181, D2:D181, "Travel", C2:C181, ">="&DATE(2026,1,1), C2:C181, "<="&DATE(20
```

> **The `">="&value` pattern:** When a criterion uses a comparison operator, write the operator as a text string and concatenate it with the value using `&`. This is required syntax — not optional.

COUNTIFS

Same pattern, but counts rows instead of summing:

```
=COUNTIFS(criteria_range1, criteria1, [criteria_range2, criteria2], ...)
```

Example — number of Credit Card transactions in Marketing:

```
=COUNTIFS(F2:F181, "Credit Card", D2:D181, "Marketing")
```

AVERAGEIFS

Same as SUMIFS but returns the average:

```
=AVERAGEIFS(average_range, criteria_range1, criteria1, ...)
```

Example — average Hardware purchase amount:

```
=AVERAGEIFS(E2:E181, D2:D181, "Hardware")
```

MAXIFS / MINIFS (Excel 2019+)

```
=MAXIFS(max_range, criteria_range1, criteria1, ...)
```

Example — largest single Training expense:

```
=MAXIFS(E2:E181, D2:D181, "Training")
```

6. Text Functions

Functions that manipulate the text content of cells — cleaning, extracting, combining.

FUNCTION	SYNTAX	WHAT IT DOES	EXAMPLE
`LEFT`	`=LEFT(text, n)`	First *n* characters	`=LEFT("E1001",1)` → `"E"`
`RIGHT`	`=RIGHT(text, n)`	Last *n* characters	`=RIGHT("E1001",4)` → `"1001"`
`MID`	`=MID(text, start, n)`	*n* chars from position *start*	`=MID("P-201",3,3)` → `"201"`
`LEN`	`=LEN(text)`	Character count	`=LEN("Priya Nair")` → `10`

FUNCTION	SYNTAX	WHAT IT DOES	EXAMPLE
`FIND`	`=FIND(find, within)`	Position of first match (case-sensitive)	`=FIND("@","a@b.com")` → `2`
`SEARCH`	`=SEARCH(find, within)`	Same as FIND, case-insensitive	`=SEARCH("NAIR","Priya Nair")` → `7`
`SUBSTITUTE`	`=SUBSTITUTE(text, old, new)`	Replaces all occurrences	`=SUBSTITUTE("P-201","P-","PRJ-")` → `"PRJ-201"`
`CONCATENATE`	`=CONCATENATE(a, b, ...)`	Joins strings	`=CONCATENATE(B2," ",C2)` → `"Priya Nair"`
`TEXTJOIN`	`=TEXTJOIN(delim, ignore_empty, ...)`	Joins with delimiter (Excel 2019+)	`=TEXTJOIN(", ",TRUE,A2:A5)`
`TEXT`	`=TEXT(value, format)`	Formats a value as text	`=TEXT(42000,"₹#,##0")` → `"₹42,000"`

Example — extract email username:

```
=LEFT(L2, FIND("@", L2)-1)     → "priya.nair"
```

Example — extract email domain:

```
=MID(L2, FIND("@", L2)+1, LEN(L2))   → "corpsolutions.com"
```

Example — full name as "Last, First":

```
=C2&", "&B2     → "Nair, Priya"
```

7. Date Functions

FUNCTION	SYNTAX	WHAT IT RETURNS	EXAMPLE
`YEAR`	`=YEAR(date)`	4-digit year	`=YEAR("12-01-2022")` → `2022`
`MONTH`	`=MONTH(date)`	Month (1–12)	`=MONTH("12-01-2022")` → `1`
`DAY`	`=DAY(date)`	Day of month	`=DAY("12-01-2022")` → `12`
`TODAY`	`=TODAY()`	Current date	Updates daily
`DATEDIF`	`=DATEDIF(start, end, unit)`	Date difference	`=DATEDIF(F2, TODAY(), "Y")` → years
`EOMONTH`	`=EOMONTH(start, months)`	End of month *n* months away	`=EOMONTH(TODAY(), 6)`
`NETWORKDAYS`	`=NETWORKDAYS(start, end)`	Working days (excl. weekends)	`=NETWORKDAYS("01-01-2026","31-03-2026")`

FUNCTION	SYNTAX	WHAT IT RETURNS	EXAMPLE
`TEXT`	`=TEXT(date, format)`	Custom date display	`=TEXT(F2, "DD-MMM-YYYY")` → `"12-Jan-2022"`

Example — years of service:

```
=DATEDIF(F2, TODAY(), "Y")
```

Example — transaction quarter:

```
="Q"&ROUNDUP(MONTH(C2)/3,0)&" "&YEAR(C2)    → "Q1 2026"
```

Example — probation end date (6 months after hire):

```
=EOMONTH(F2, 6)
```

Example — working days in Q1 2026:

```
=NETWORKDAYS(DATE(2026,1,1), DATE(2026,3,31))    → 64
```

8. Error Handling

Errors are signals, not bugs — but in a report, raw `#N/A` or `#DIV/0!` looks broken. Error-handling functions intercept them.

Common errors

ERROR	TYPICAL CAUSE
`#N/A`	Lookup can't find the value
`#VALUE!`	Wrong data type in a formula
`#REF!`	Referenced cell was deleted
`#DIV/0!`	Division by zero
`#NAME?`	Unrecognised function name

IFERROR — catches any error

```
=IFERROR(formula, value_if_error)
```

```
=IFERROR(VLOOKUP(B2, employee_master!$A$2:$N$41, 4, FALSE), "Unknown Dept")
```

IFNA — catches only #N/A

```
=IFNA(VLOOKUP(B2, employee_master!$A$2:$N$41, 4, FALSE), "ID Not Found")
```

Best practice: Use `IFNA` for lookups (the expected failure is `#N/A`). Use `IFERROR` only when you genuinely want to suppress *all* error types.

ISERROR / ISNA — for conditional branching

Returns `TRUE` / `FALSE` — useful inside `IF` when you want to *decide* based on an error, not just suppress it:

```
=IF(ISNA(VLOOKUP(B2, ..., 4, FALSE)), "Invalid ID", "OK")
```

Warning: Don't wrap everything in `IFERROR(..., "")` — this hides genuine errors that should be investigated. Only suppress errors you've thought through.

Break (10 min)

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9. PivotTables & PivotCharts

Definition: A **PivotTable** is an interactive summary of a dataset. You drag fields into four areas to instantly group, aggregate, and filter data — no formulas needed.

The four areas

AREA	WHAT GOES HERE	EXAMPLE
Rows	Categories down the left side	Department, Category
Columns	Categories across the top	Payment Mode, Quarter
Values	Numbers being summarised	Sum of Amount, Count of Txn ID
Filters	Fields that filter the entire PivotTable	Project Code, Region

Building a PivotTable

1. Click anywhere inside your data (or select the full range)
2. **Insert** → **PivotTable** → choose "New Worksheet"
3. In the **PivotTable Field List** pane on the right, drag fields:
4. Drag **Category** to Rows
5. Drag **Amount** to Values (defaults to Sum)
6. Result: total Amount by Category — one row per category, summed automatically

Making it more useful

Cross-tab: Drag **Payment Mode** to Columns → each payment method becomes a column, showing spend by Category × Payment Mode.

Change aggregation: Click the dropdown arrow on "Sum of Amount" in Values → **Value Field Settings** → choose Count, Average, Max, etc.

Date grouping (Excel 365): Drag a Date field to Rows → right-click any date → **Group** → choose Months, Quarters, or Years. No helper column needed.

Show Values As: Right-click a value cell → **Show Values As** → choose **% of Grand Total**, **% of Row Total**, **% of Column Total**, or **Running Total**. Turns raw numbers into proportions.

Sort: Click any value cell → sort descending to see the highest-spend categories first.

Filter: Use dropdown arrows on Row/Column labels. Or drag a field to the Filters area for a report-level filter.

Refresh: When source data changes, right-click the PivotTable → **Refresh**. If you created the PivotTable from a Table (`Ctrl+T`), new rows are included automatically on refresh.

PivotCharts

1. Click inside the PivotTable
2. **PivotTable Analyze** → **PivotChart** (or **Insert** → **PivotChart**)
3. Choose a chart type: Column, Bar, Pie, Line
4. The chart has the same filter buttons — clicking a filter updates both chart and table
5. Rearranging fields in the PivotTable automatically updates the chart

Example workflow: 1. PivotTable: Rows = Category, Values = Sum of Amount → insert a Clustered Column chart 2. Add Payment Mode to Columns → chart updates to grouped columns 3. Filter to "Credit Card" and "Bank Transfer" only → chart and table filter together 4. Sort by Sum descending → chart reorders

Drill-down: Double-click any value cell in a PivotTable — Excel creates a new sheet with just the underlying rows. Useful for investigating outliers.

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10. Exercises

Exercise 1: Lookup Drills

Add these columns to `monthly_transactions.csv` by looking up each Emp ID in `employee_master.csv`:

NEW COLUMN	FUNCTION TO USE	RETURN COLUMN
Employee Name	XLOOKUP (or VLOOKUP + concatenation)	First Name & Last Name
Department	XLOOKUP or INDEX-MATCH	Department
Designation	INDEX-MATCH	Designation
Hire Date	XLOOKUP	Hire Date
Base Salary	VLOOKUP	Base Salary
Region	INDEX-MATCH	Region
Manager	INDEX-MATCH	Manager

NEW COLUMN	FUNCTION TO USE	RETURN COLUMN
Performance Rating	XLOOKUP	Performance Rating

Then answer: a) Which employee (by name) has the most transactions? b) What is the average transaction amount for employees rated 5? c) Are there transactions by "On Leave" or "Resigned" employees?

Exercise 2: Reverse and Approximate Lookups

a) Given a Manager name, find the first Emp ID they manage. Which lookup method works?

b) Create this tax-bracket table and use VLOOKUP (TRUE) or XLOOKUP (match_mode -1) to find each employee's tax rate:

INCOME SLAB	TAX RATE
0	0%
30000	5%
50000	10%
75000	15%
100000	20%

Exercise 3: SUMIFS / COUNTIFS / AVERAGEIFS Dashboard

Build these metrics — each with a single formula, no filtering:

#	METRIC
1	Total spend on Travel
2	Total Marketing spend for Project P-403
3	Number of Credit Card transactions
4	Transactions approved by "Fatima Sheikh" in Q1 (Jan–Mar 2026)
5	Average Hardware purchase amount
6	Total Finance department spend in Q2 (Apr–Jun 2026)
7	Petty Cash transactions under ₹3,000
8	Total Training spend in the North region
9	Transactions where Amount > ₹30,000
10	Average Software License cost paid by Bank Transfer

Exercise 4: Text Functions

Using `employee_master.csv`:

a) **Full Name:** `=C2&", "&B2` → `"Nair, Priya"`

b) **Email username:** `=LEFT(L2, FIND("@",L2)-1)` → `"priya.nair"`

c) **Email domain:** `=MID(L2, FIND("@",L2)+1, LEN(L2))` → `"corpsolutions.com"`

d) **Masked phone:** `="XXXXXX"&RIGHT(M2,4)` → `"XXXXXX3210"`

e) **Standardise "Sr." to "Senior":** `=SUBSTITUTE(E2, "Sr.", "Senior")`

f) **Project number from code:** `=MID(I2,3,3)` on `"P-201"` → `"201"`

g) **Narrative sentence:** `=B2&" "&C2&" from "&D2&" spent ₹"&TEXT(E2,"#,##0")&" on "&D2` (adapt columns as needed)

Exercise 5: Date Functions

a) **Years of service:** `=DATEDIF([@[Hire Date]], TODAY(), "Y")`

b) **Tenure bucket:** `=IF(DATEDIF(F2,TODAY(),"Y")>=5, "5+ years", IF(DATEDIF(F2,TODAY(),"Y")>=2, "2–5 years", "< 2 years"))`

c) **Hire quarter:** `="Q"&ROUNDUP(MONTH(F2)/3,0)&" "&YEAR(F2)`

d) **Transaction quarter:** Same formula on the Transaction Date column

e) **Probation end:** `=EOMONTH(F2, 6)`

f) **Working days in Q1 2026:** `=NETWORKDAYS(DATE(2026,1,1), DATE(2026,3,31))`

g) **Formatted label:** `=TEXT(F2, "DD-MMM-YYYY")` → `"12-Jan-2022"`

Exercise 6: Error Handling

a) Change one Emp ID to `"E9999"`. Observe the `#N/A`. Wrap in `IFNA` with `"ID Not Found"`.

b) Calculate Average Spend per Category = `SUMIFS / COUNTIFS`. Handle `#DIV/0!` with `IFERROR(..., 0)`.

c) Create a validation column: `=IF(ISNA(VLOOKUP(B2,...)), "Invalid ID", "OK")`

Exercise 7: PivotTable & PivotChart

a) PivotTable: Rows = Category, Values = Sum of Amount. Which category spends the most?

b) Add Department to Rows, Category to Columns → cross-tab. Which dept-category pair dominates?

c) Add a Month column → Rows = Month, Columns = Category. Any spending trends?

d) Rows = Approved By, Values = Sum of Amount + Count of Txn ID. Who approves the most? The highest value?

e) Filter by Project Code = P-102. Which categories dominate?

f) Insert a Clustered Column PivotChart from the Category PivotTable. Add Payment Mode to Columns. Filter to Credit Card + Bank Transfer.

g) Build a Line PivotChart: Rows = Month, Values = Sum of Amount. Add a trendline. Trend direction?

h) Double-click the highest value cell — explore the drill-down detail sheet.

11. Exit Ticket

1. Write a VLOOKUP that finds `"E1015"` in the employee master and returns the Department. What changes for Base Salary?
2. Rewrite it using INDEX-MATCH. Why is INDEX-MATCH more robust when columns are inserted?
3. Write a SUMIFS for total "Software License" spend paid by "Bank Transfer".
4. Write a formula to extract the email username (before `@`) from cell L2.
5. A VLOOKUP returns `#N/A` for a new employee. Wrap it in an error handler. Would you use IFERROR or IFNA — and why?

Looking ahead: Today's lookups, aggregation, and PivotTables are the analytical core — next session covers data validation, conditional formatting, what-if analysis, and dashboard design.